

**STRULE AUTOMATION**

Industrial Controls · Bay Area & Northern California

TRAINING PRIMER

Reading Electrical and Control Drawings

I have watched a good mechanic stand in front of an open control panel with a two-inch stack of drawings and not know which page tells him where a wire lands. That is not a knock on him. Nobody ever sat him down and explained what the different sheets are *for*. Once you know that, the stack stops being intimidating and starts being what it actually is: a map. These are my field notes on how to read that map, the way I'd walk a sharp maintenance tech through it standing at the panel.

Why this matters on the floor

When a conveyor won't start and the operator is leaning on you, the drawings are the fastest honest answer you have. They tell you what the designer *intended* the circuit to do. They will not tell you whether somebody rewired it in 2019 and never marked it up — we'll get to that hard truth — but they give you a place to start that isn't guessing. A tech who can trace a circuit across two or three sheets can find a bad start permissive in ten minutes. A tech who can't will pull contactors and swap parts for an hour and maybe get lucky.

The catch is that "the drawings" are not one document. They're several different kinds of drawing, each drawn to answer a different question. Read them expecting the wrong answer and they feel like nonsense. So let's take them one at a time.

The P&ID: what the process is trying to do

The **P&ID (piping and instrumentation diagram)** is the big-picture process drawing. It shows you the tanks, pumps, conveyors, valves, and the pipes or material paths between them, plus every instrument that measures or controls something along the way. It does *not* show wiring and it does *not* show control logic. It answers one question: what is the process, and what is watching it?

On an aggregate or organics plant, a P&ID might show a feed hopper, a belt to a screen, a sump with a level element, and a pump that empties the sump. Each instrument sits on the drawing as a little circle — a "balloon" — with a tag inside it like **LT-205** or **FV-110**. The lines between equipment are the process (material, slurry, water, biogas), and dashed or special lines show signal connections. You read a P&ID to understand *what talks to what*, before you ever open a panel.

You don't trace wires on a P&ID. You use it to learn the vocabulary of the plant — the tag names — so that when you see the same tag on an electrical sheet you know which physical thing in the field it points to.

ISA-style instrument tags: reading the balloon

Those tags follow a convention worth learning, because once you know it you can read an instrument you've never seen before. The common standard is from the **ISA (International Society of Automation)**, and at a basic level a tag is **letters plus a loop number**, like **FT-101**.

The letters are a little code. The **first letter** tells you the measured or initiating variable — what the instrument is about:

- **F** = flow
- **L** = level
- **P** = pressure
- **T** = temperature
- **A** = analysis (for example, a gas concentration)
- **S** = speed

The **following letters** tell you what the device *does* with that variable:

- **T** = transmitter (sends a signal)
- **I** = indicator (shows a reading)
- **C** = controller (acts on it)
- **V** = valve (the final control element)
- **E** = element (the raw sensor)
- **S** = switch

So **FT-101** reads as *flow transmitter, loop 101*. **LT-205** is a *level transmitter, loop 205*. **PIC-110** is a *pressure indicating controller, loop 110*. **PV-110** is the *pressure control valve* — the final element that controller drives — in that same loop 110, and the shared loop number ties them together as one control loop. (Notice the valve carries the loop's variable letter, P, not the letter of whatever it happens to throttle; the final element takes the first letter of the loop it belongs to.) **TE-301** is the *temperature element*, the actual probe, and **TT-301** is the transmitter that turns the probe's signal into something the control system can read.

That loop number is the thread that stitches a loop together across every drawing. Everything sharing loop 110 is part of the same measure-and-act story. Check your own plant's tag legend, because sites bend the convention, but the ISA pattern underneath is remarkably consistent.

The schematic: ladder logic and rungs

Now we move from *what the process is* to *how the control decides*. That's the **electrical schematic**, also called an **elementary diagram** or **ladder diagram**. The name "ladder" is literal: two vertical lines are the power rails, like the sides of a ladder, and each horizontal **rung** between them is one piece of control logic.

Read a rung left to right as a sentence. On the left rail is the source of control power. As you move right you pass through **contacts** — little symbols that are either open or closed — and each contact is a condition. At the right end sits a **coil** or a **load**: the thing that turns on if every condition to its left is satisfied. Two vertical parallel lines is a normally-open contact; the same with a slash is normally-closed. A circle at the end is a relay coil; it might energize a **motor starter** or a control relay.

So a rung might read, left to right: control power, then an E-stop contact (closed when the E-stop is not pushed), then an overload contact (closed when the motor's overload hasn't tripped), then a start pushbutton in parallel with a holding contact, and finally the starter coil. In plain words: *if the E-stop is clear and the overload is fine and somebody pressed start, energize the starter, and the holding contact keeps it energized after they let go*. That's a standard motor start/stop rung, and once you see one you'll see it everywhere.

Every one of those conditions to the left of the coil is a **permissive** or an **interlock** — something that must be true before the motor is allowed to run. This is exactly where people get tempted to cheat, and it's where I have to be blunt: those interlocks are there on purpose. A guard-closed contact, a gas-detection permissive, a high-level shutdown — you do not jumper those out to make a machine run. If a permissive is a genuine nuisance trip and the setpoint is wrong, the legitimate fix is a management-of-change: one documented adjustment, inside the equipment's rated limit, logged, with the right people signing off. Never defeat a protective function to get through a shift.

The wiring diagram and panel layout: where wires actually land

The schematic tells you the *logic*. It draws that E-stop contact and that coil in whatever position makes the logic easy to read — which is almost never where they physically sit. To find the actual wire, you need two more drawings.

The **wiring diagram** (or **connection diagram**) shows where wires physically land: which terminal, which terminal block, which point on the starter. It's the drawing that says "wire number 14 lands on terminal block TB2, point 7, and the other end lands on the starter coil terminal A1." It is less pretty than the schematic and far more useful when you have a screwdriver in your hand.

The **panel layout** is the drawing of the inside of the enclosure, roughly to scale, showing where each component physically sits — this relay here, that terminal strip along the bottom, the PLC in the upper left. When the wiring diagram says "K3," the panel layout tells you *which physical relay* K3 is so you're not staring at eight identical relays.

To make the two connect, wires are numbered. **Wire numbering** means every conductor carries a printed tag — a ferrule or a heat-shrink label — with a number that matches the drawings. The convention varies, but a common one is that a wire keeps the same number along its whole run, and the number often ties to the rung or the signal it carries. Find wire 14 on the schematic, and you can find wire 14 printed on the actual conductor in the panel. That is the bridge between the paper and the copper.

■ The loop sheet: one instrument loop, all the way through

For an individual instrument, there's often a dedicated **loop sheet** (loop diagram). Where the schematic shows control logic and the wiring diagram shows the panel, the loop sheet follows *one* measurement loop from the field device all the way back to the control system: the transmitter in the field, the wire pair, the junction box, the terminal in the marshalling cabinet, the input card and channel on the **PLC (programmable logic controller)**. If **LT-205** is reading wrong, the loop sheet tells you every terminal that signal passes through between the tank and the processor. It's the single most efficient drawing for troubleshooting a dead or drifting instrument.

■ Cross-references: how the sheets point at each other

Here's the piece that unlocks the whole stack. A relay coil energized on one rung usually has its *contacts* used on other rungs, often on other sheets. You can't draw a wire across three pages, so instead the drawings use **cross-references**.

Next to a coil you'll see a list of numbers — those point to every place that relay's contacts appear. A common format is sheet-and-rung, like **12/4**, meaning "this coil's contact is used on sheet 12, rung 4." Next to each contact, in turn, is a reference back to where its coil lives. So a coil on sheet 8 might list **12/4** and **15/2**, and when you flip to sheet 12 rung 4 you find that contact with a little note pointing back to sheet 8. Follow the references and you can walk a signal across the whole drawing set without ever losing the thread.

Sheet-to-sheet references work the same way for wires that leave one page and continue on another: an arrow at the edge of the sheet with a code telling you which sheet and zone it continues on. The drawing is broken into a grid — letters down the side, numbers across the top — so a reference like "sheet 15, C-4" sends you to a specific box on a specific page.

■ Worked example: tracing a motor start end to end

Let's trace one, the way I'd do it at a panel. Say the operator reports that conveyor **CV-120** won't start.

I begin at the **P&ID** to confirm what CV-120 is and what's tied to it — is there a level or a jam interlock upstream that could be holding it? Good; I know the players.

I go to the **schematic** and find the start/stop rung for CV-120's starter, call the coil **M120**. I read the rung left to right and check each permissive: E-stop, motor overload, a downstream-running interlock, the start pushbutton and its holding contact. One contact in that rung is a cross-reference — it's the "downstream conveyor running" permissive, shown as **CR15** with a note 22/3. That tells me the contact belongs to relay CR15, whose coil lives on sheet 22, rung 3.

I flip to **sheet 22, rung 3** and read what makes CR15 pick up. Suppose CR15 only energizes when the downstream belt's run feedback is present — and that feedback is missing. Now I know *why* CV-120 won't start: it's correctly refusing because its downstream unit isn't confirmed running. The logic is doing its job.

Now I want to physically check that feedback wire. I note its **wire number** off the schematic — say wire 2214. I go to the **wiring diagram** to see where 2214 lands: terminal block TB4, point 12, wired to a PLC input card. The **panel layout** shows me where TB4 physically sits inside the enclosure. And before I touch anything, I confirm the state with a meter — after I've made the circuit safe, which is the next section and the whole point.

That's the full path: field tag on the P&ID, to the rung on the schematic, across a cross-reference to another sheet, to a wire number, to the terminal on the wiring diagram, to the physical spot on the panel layout. Each drawing answered the single question it was drawn for, and in sequence they carried me from the field tag to the exact terminal I needed to check.

■ The title block, revisions, and red-line discipline

Every sheet has a **title block**, usually bottom-right: drawing number, sheet number, title, the plant, and a **revision** block. The revision tells you *which version* you're holding. Two people troubleshooting from two different revisions of the same sheet will argue all afternoon and both be right about their own paper. Always check that everyone's on the same rev.

When a wire actually gets changed in the field, the right discipline is a **red-line**: you mark the change in red on a controlled copy of the drawing, date it, initial it, and get it back to whoever maintains the master so the next revision captures it. Red-lining is how a drawing set stays honest. Skipping it is how it rots.

■ The hard truth: the print is not the plant

Which brings me to the thing that matters most. Drawings drift from reality. Somebody adds a jumper at 2 a.m. to keep a line running, a motor gets swapped for a different frame, a relay gets replaced with whatever was on the shelf and wired a little differently. If nobody red-lined it, the print now lies — not out of malice, just entropy.

So treat the drawing as a strong hypothesis, never as proof. And understand this clearly for your own safety: **a print showing a circuit de-energized is not proof that the circuit is dead**. The drawing shows intent; the copper shows reality, and only a meter tells you which one you've got.

Before you put hands on any conductor, do **LOTO (lockout/tagout)**: bring the equipment to a zero-energy state, lock the disconnect, and tag it. Then **test before touch** — verify de-energized with a meter you've proven works on a known live source first, then check your circuit, then re-check your meter on the known source. When you're taking a *live* reading because a measurement genuinely requires it, that's energized work: rated gloves and PPE, an arc-flash assessment, and the awareness that you're exposed. On anything with a variable frequency drive, respect the **stored DC-bus energy** — the drive holds a lethal charge for minutes after power-down, so wait out the rated discharge time and verify before opening it. The drawing helps you plan the work. It never makes the circuit safe. Your meter and your locks do that.

Try this on your own plant (safely)

Pick one motor you know well and, with the plant running normally and the panel *closed*, find its start/stop rung on the schematic. Read the rung out loud as a sentence and name every permissive. Then find one cross-referenced contact and flip to the sheet it points to. You've now traced a circuit across sheets without ever opening a live enclosure — pure paper practice, zero exposure. Do that with three or four circuits and the whole stack stops being a stack and starts being a map you can read.

Glossary

- **P&ID (piping and instrumentation diagram)**: the process-level drawing showing equipment, piping/material paths, and instruments with their tags; shows what the process does, not the wiring.
- **ISA (International Society of Automation)**: the body behind the common instrument-tagging convention.
- **Instrument tag**: letters plus a loop number (e.g., FT-101); first letter = measured variable, following letters = device function.
- **Schematic / elementary / ladder diagram**: the control-logic drawing, drawn as rungs between two power rails.
- **Rung**: one horizontal line of a ladder diagram; one piece of control logic read left to right.
- **Contact**: a switch symbol on a rung representing a condition (normally-open or normally-closed).
- **Coil**: the load at the end of a rung, often a relay or motor starter, that energizes when the rung's conditions are met.
- **Permissive / interlock**: a condition that must be true before an action is allowed; a protective function you never defeat.
- **Motor starter**: the contactor-and-overload device that switches power to a motor.
- **Wiring / connection diagram**: the drawing showing where wires physically land — terminals and blocks.

- **Panel layout:** the scale drawing of the enclosure interior showing where each component physically sits.
- **Wire numbering:** the printed tag on each conductor that matches the drawings, bridging paper to copper.
- **Loop sheet (loop diagram):** the drawing following one instrument loop from field device through every terminal to the PLC.
- **PLC (programmable logic controller):** the industrial processor that runs the control logic.
- **Cross-reference:** the sheet/rung note beside a coil or contact pointing to where its related contacts or coil appear.
- **Title block:** the bottom-corner block with drawing number, sheet, title, and revision.
- **Revision:** the version identifier of a drawing; confirm everyone is on the same one.
- **Red-line:** a dated, initialed field markup of a drawing change, fed back to update the master.
- **LOTO (lockout/tagout):** bringing equipment to zero-energy state and locking/tagging it before hands-on work.
- **Test before touch:** verify de-energized with a meter proven on a known live source before and after.
- **DC-bus energy:** the lethal stored charge a variable frequency drive holds after power-down until it discharges.

A last note, kept short and on-topic: the drawings drift most at the seams — where one vendor's panel got handed to the next vendor's integration, where a control upgrade landed on top of legacy wiring. That's where a second set of eyes earns its keep, and it's usually where the print and the plant have quietly stopped agreeing.